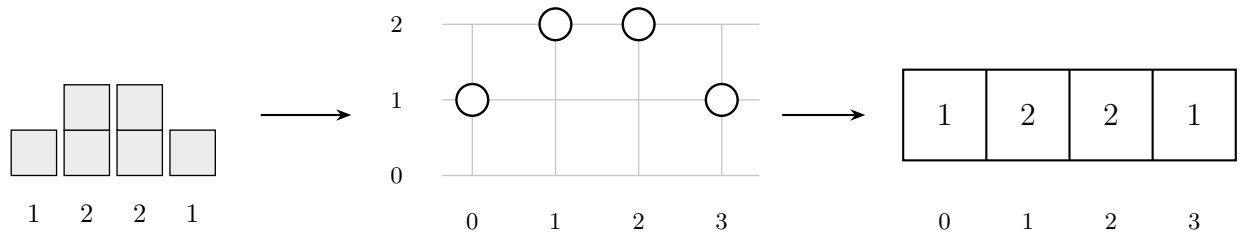
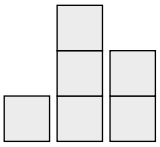
Heights are whole numbers, including 0. Neighbors differ by at most 1. Black circles and shaded boxes are fixed clues.

One marker in each column shows its tower height.

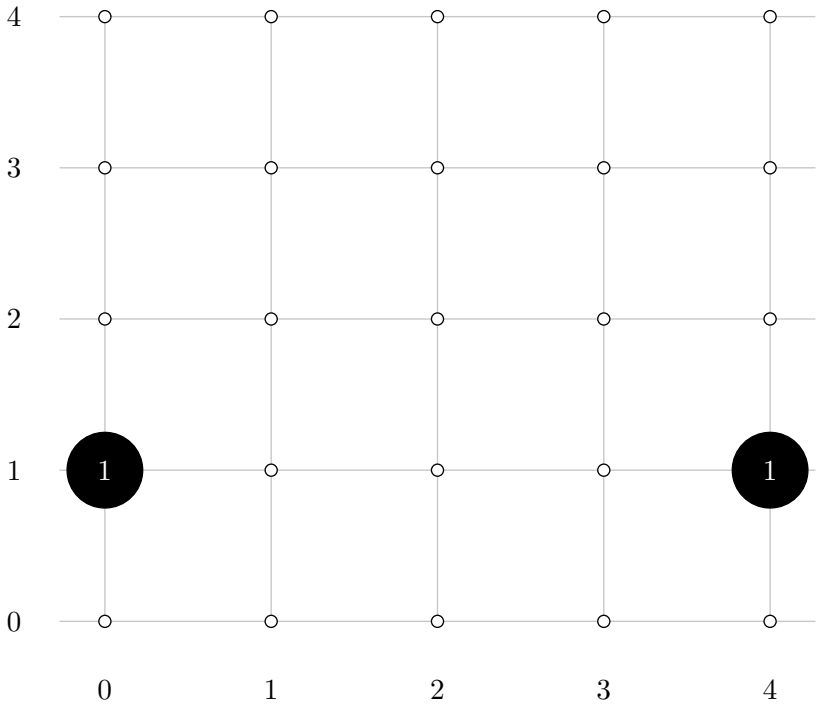


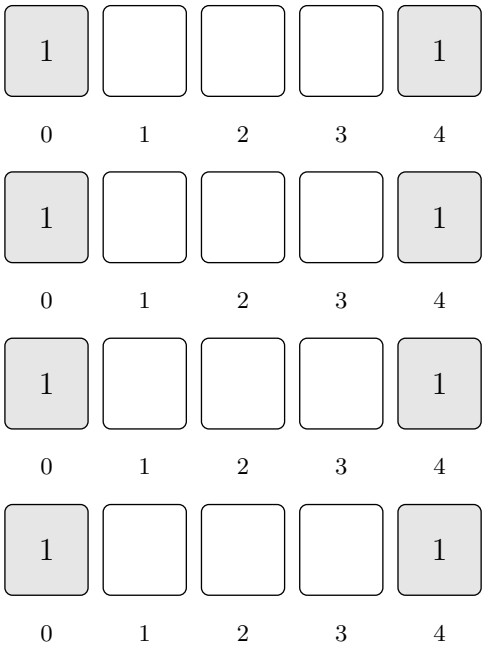
1 2 2 1
allowed



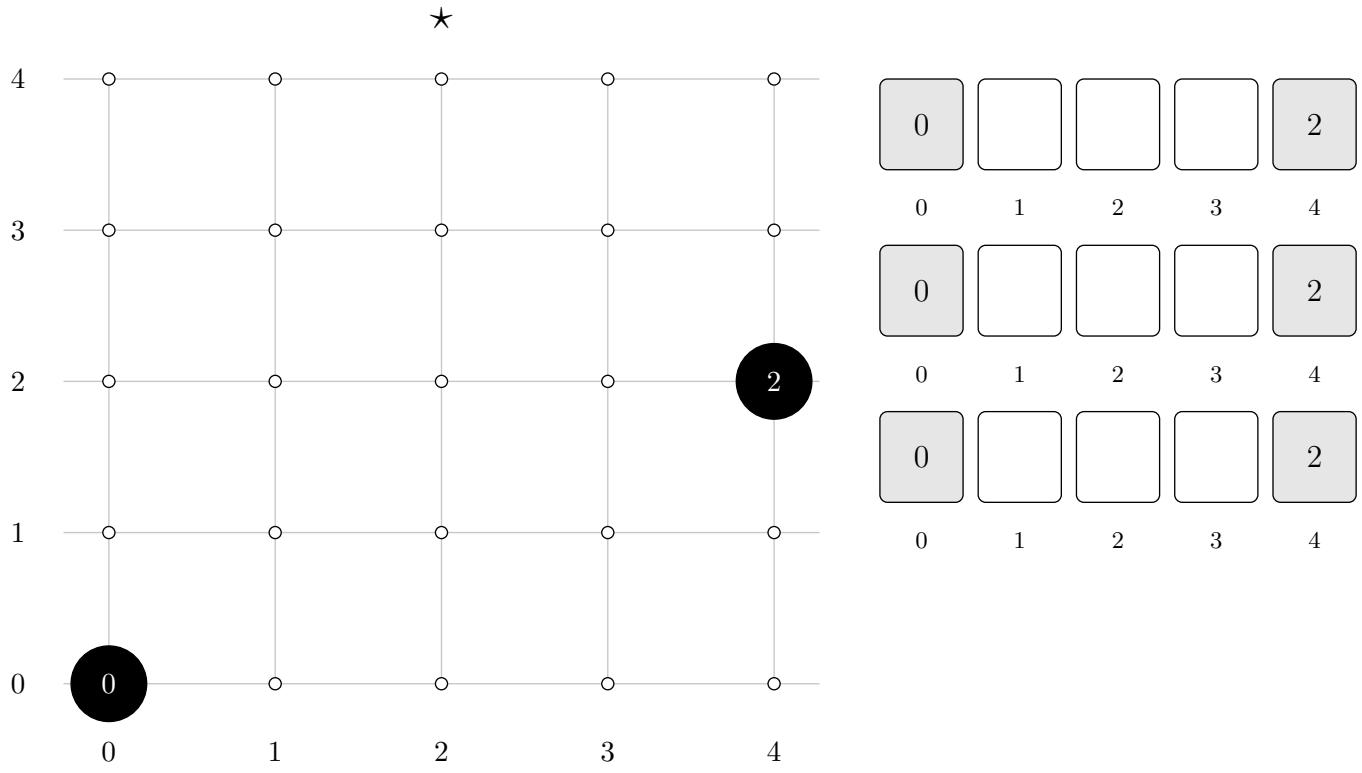
1 3 2
jump too big

Problem 1: Make four different landscapes with these clues. Can the middle height be 0? Can it be 3?

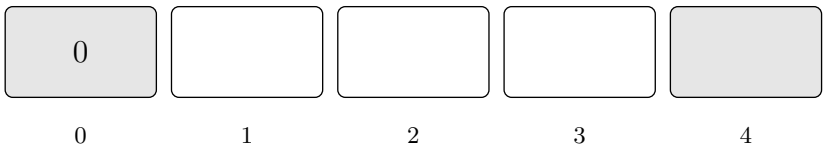




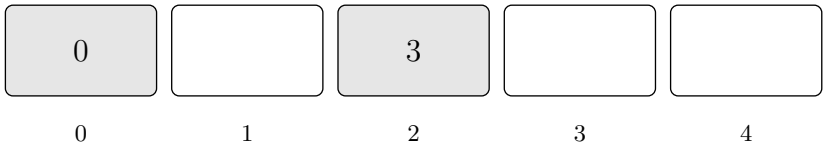
Problem 2: Which heights can the starred marker have? Make a landscape for each possible height and explain why the other heights cannot work.

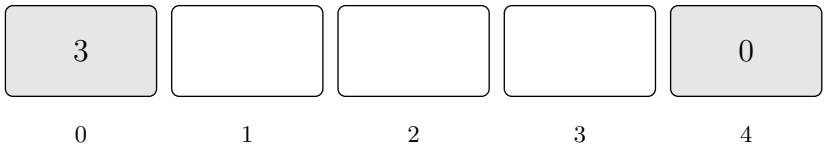


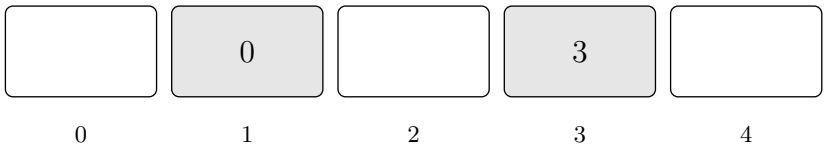
Problem 3: Keep the left clue at 0. Choose a height for the right clue that forces every marker. Find all such right-clue heights.

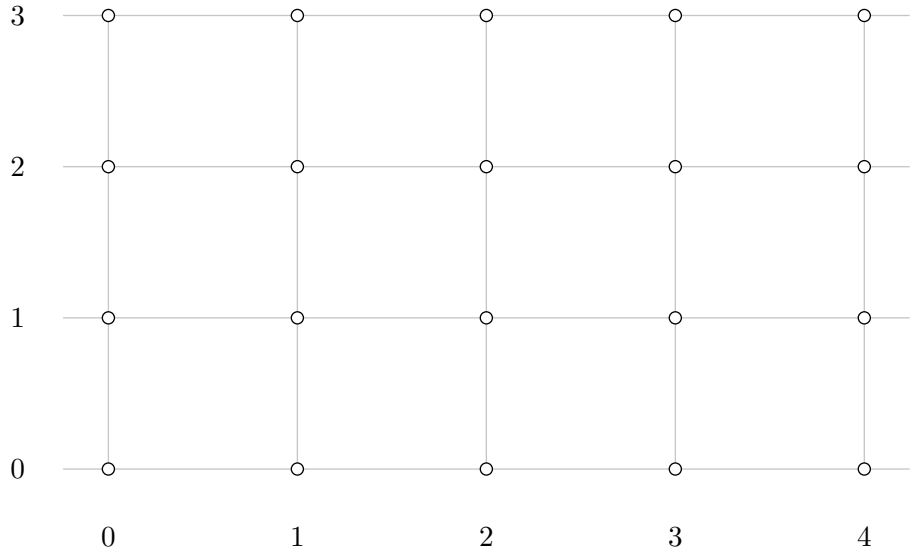


Problem 4: Decide which clue sets can be completed. Build the possible ones; explain what prevents the others.

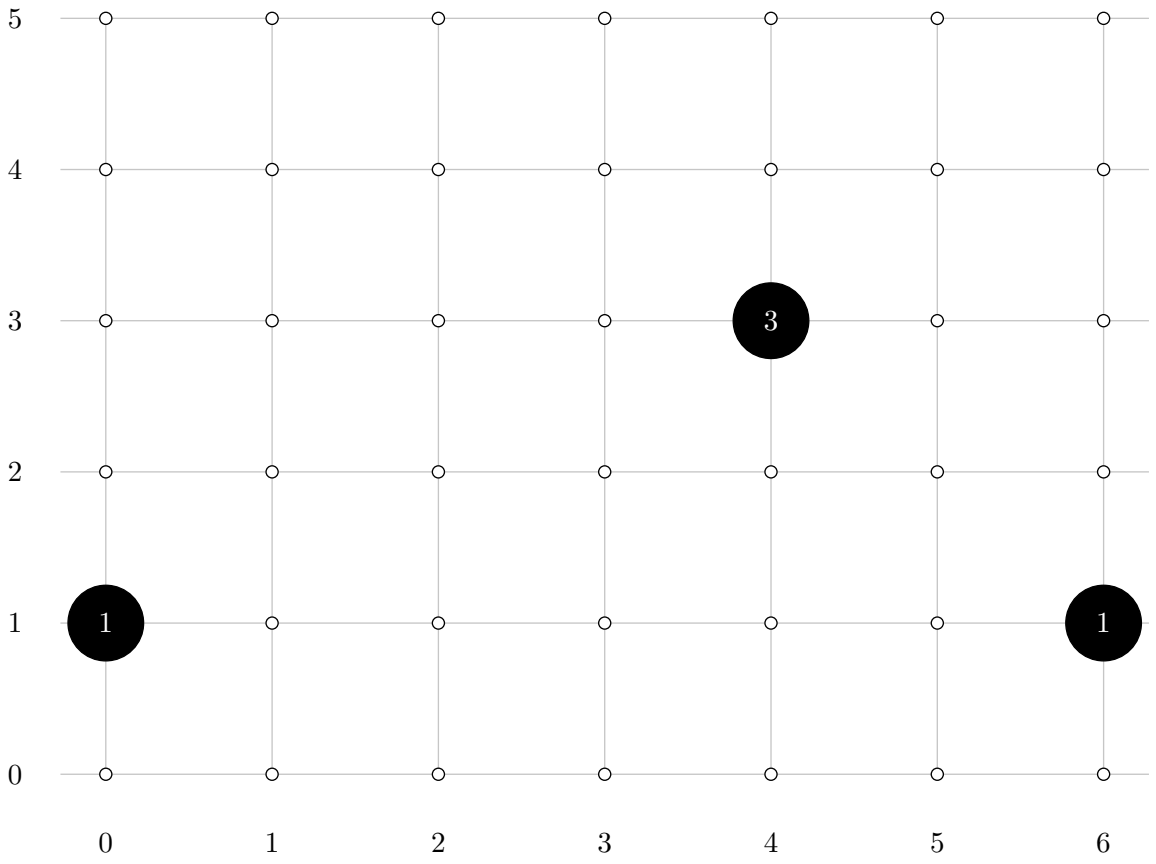


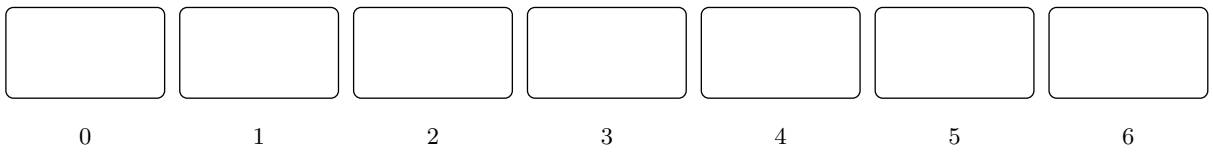


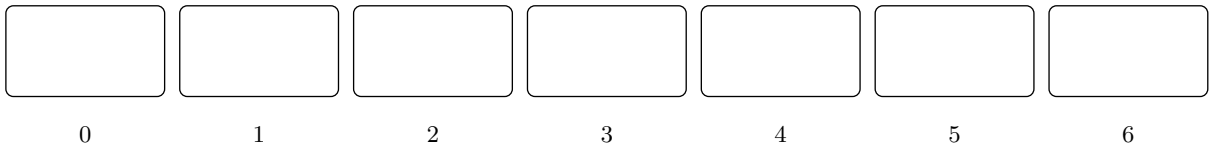




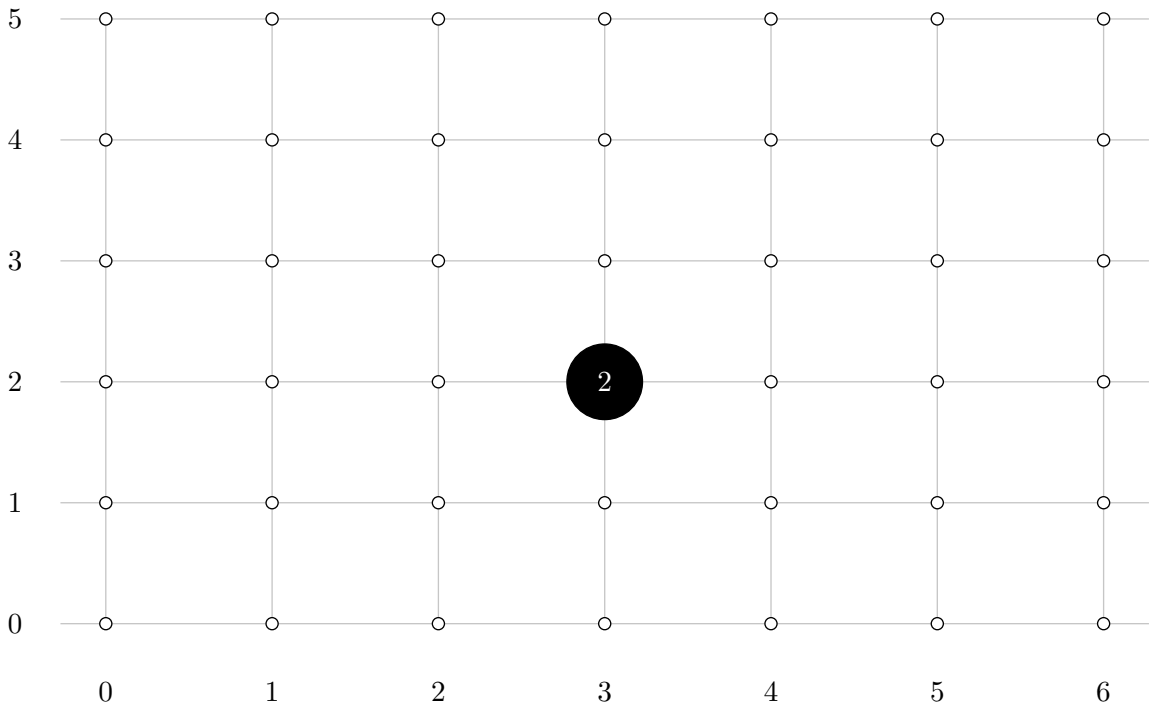
Problem 5: Is there a landscape that is as low as possible at every site at once? One that is as high as possible at every site? Build any that exist, and explain why no marker can go beyond them.

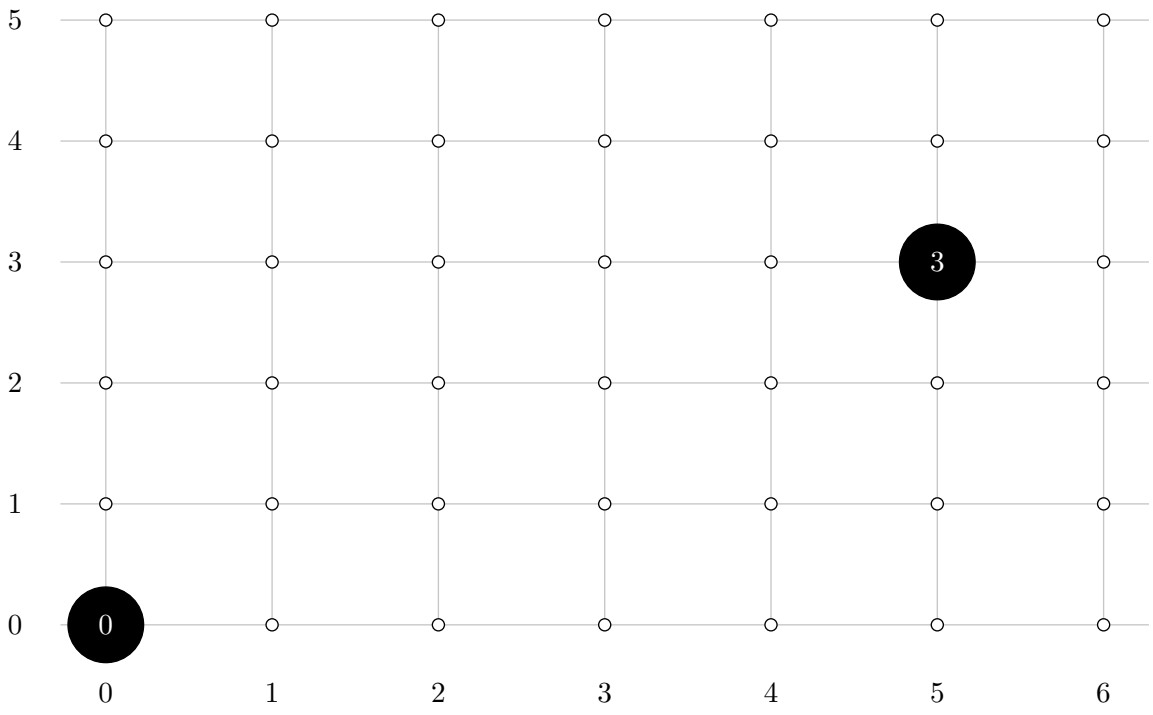






Problem 6: On each board, shade every height each white marker can have. If you pick one shaded height in every column, do your picks always make a landscape?





Problem 7: A row has at least one clue. Suppose every pair of clues differs by no more than the steps between them. Must the whole row be fillable? Give a construction that always works, or a set of clues that defeats the claim.

